Answer key

Image 1 is the fifth matrix; Image 2 is the sixth. The numbers identify the top (1), middle (2) or bottom (3) response in each choice column.

Exercise	I1	I2	Exercise	I1	I2	Exercise	I1	I2	Exercise	I1	I2
1	3	2	36	2	1	71	1	2	106	1	1
2	2	1	37	2	2	72	3	1	107	3	2
3	3	3	38	1	2	73	1	3	108	3	2
4	1	2	39	3	1	74	2	3	109	2	1
5	2	1	40	2	2	75	3	1	110	1	1
6	3	3	41	2	3	76	1	3	111	1	2
7	2	1	42	3	2	77	2	2	112	2	3
8	2	2	43	1	3	78	3	3	113	3	3
9	1	2	44	2	1	79	1	3	114	3	3
10	3	2	45	1	1	80	3	1	115	2	1
11	1	3	46	1	3	81	2	1	116	2	1
12	2	3	47	1	2	82	2	2	117	2	3
13	1	1	48	1	3	83	3	1	118	3	1
14	3	3	49	3	2	84	2	2	119	1	3
15	1	1	50	3	3	85	1	1	120	1	2
16	2	2	51	1	2	86	2	3	121	3	3
17	1	3	52	2	1	87	1	3	122	2	2
18	2	2	53	3	1	88	1	1	123	3	2
19	3	2	54	2	2	89	1	1	124	1	3
20	3	3	55	3	2	90	2	1	125	2	2
21	3	3	56	2	2	91	1	2	126	2	3
22	2	1	57	2	1	92	1	1	127	3	2
23	3	1	58	2	1	93	1	2	128	3	1
24	1	3	59	3	2	94	2	2	129	1	2
25	1	1	60	1	2	95	2	3	130	3	2
26	1	1	61	3	3	96	2	1	131	2	1
27	2	2	62	2	3	97	1	3	132	2	1
28	2	1	63	3	3	98	3	2	133	1	2
29	3	3	64	3	3	99	1	1	134	3	3
30	3	3	65	3	2	100	3	1	135	1	3
31	1	2	66	3	1	101	3	1	136	1	3
32	3	1	67	1	3	102	1	3	137	1	3
33	2	3	68	1	3	103	2	1	138	3	1
34	1	2	69	2	1	104	1	2	139	1	1
35	3	1	70	2	2	105	2	2	140	2	2

Solution - Exercise 1

Image 1
Matrix 3

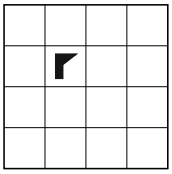
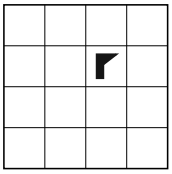


Image 2
Matrix 2



The flag (initially black) moves along row 2 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Flag: R2C2

Image 2 positions
Flag: R2C3

Solution - Exercise 2

Image 1
Matrix 2

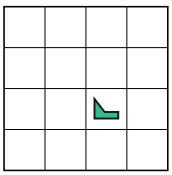
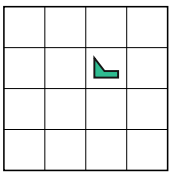


Image 2
Matrix 1



The flag (initially green) moves along column 3 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Flag: R3C3

Image 2 positions
Flag: R2C3

Solution - Exercise 3

Image 1
Matrix 3

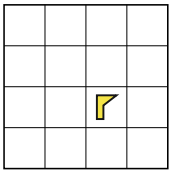
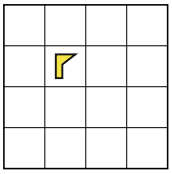


Image 2
Matrix 3



The flag (initially yellow) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Flag: R3C3

Image 2 positions
Flag: R2C2

Solution - Exercise 4

Image 1
Matrix 1

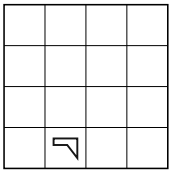
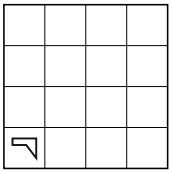


Image 2
Matrix 2



The flag (initially white) moves clockwise around the outer edge, by 1 cell per transition.

Image 1 positions
Flag: R4C2

Image 2 positions
Flag: R4C1

Solution - Exercise 5

Image 1
Matrix 2

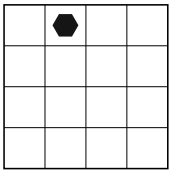
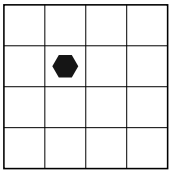


Image 2
Matrix 1



The hexagon (initially black) moves anticlockwise around the four cells R1C2, R1C3, R2C3, R2C2, by 1 cell per transition.

Image 1 positions
Hexagon: R1C2

Image 2 positions
Hexagon: R2C2

Solution - Exercise 6

Image 1
Matrix 3

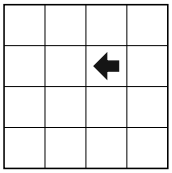
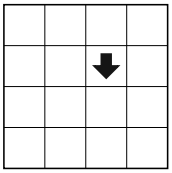


Image 2
Matrix 3



The arrow (initially black) stays at R2C3; turns 90 degrees anticlockwise each time.

Image 1 positions
Arrow: R2C3

Image 2 positions
Arrow: R2C3

Solution - Exercise 7

Image 1
Matrix 2

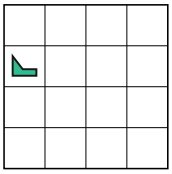


Image 2
Matrix 1

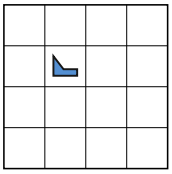


Image 1 positions
Flag: R2C1

Image 2 positions
Flag: R2C2

The flag (initially green) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle green, blue.

Solution - Exercise 8

Image 1
Matrix 2

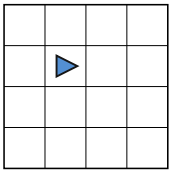


Image 2
Matrix 2

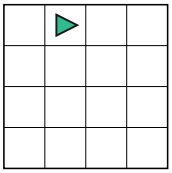


Image 1 positions
Triangle: R2C2

Image 2 positions
Triangle: R1C2

The triangle (initially blue) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle blue, green.

Solution - Exercise 9

Image 1
Matrix 1

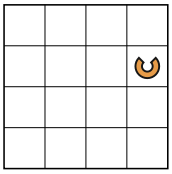
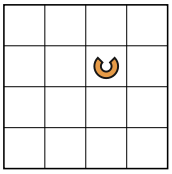


Image 2
Matrix 2



The crescent (initially orange) moves along row 2 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Crescent: R2C4

Image 2 positions
Crescent: R2C3

Solution - Exercise 10

Image 1
Matrix 3

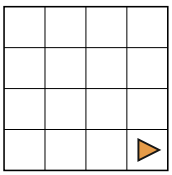
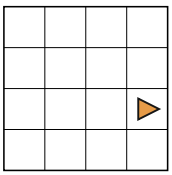


Image 2
Matrix 2



The triangle (initially orange) moves along column 4 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Triangle: R4C4

Image 2 positions
Triangle: R3C4

Solution - Exercise 11

Image 1
Matrix 1

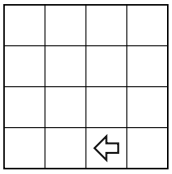
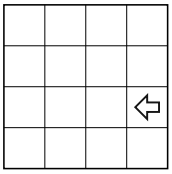


Image 2
Matrix 3



The arrow (initially white) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Arrow: R4C3

Image 2 positions
Arrow: R3C4

Solution - Exercise 12

Image 1
Matrix 2

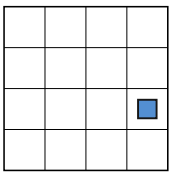
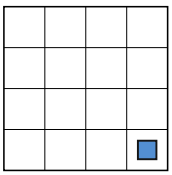


Image 2
Matrix 3



The square (initially blue) moves clockwise around the outer edge, by 1 cell per transition.

Image 1 positions
Square: R3C4

Image 2 positions
Square: R4C4

Solution - Exercise 13

Image 1
Matrix 1

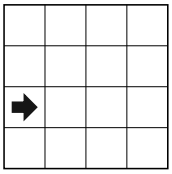
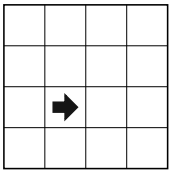


Image 2
Matrix 1



The arrow (initially black) moves anticlockwise around the four cells R2C1, R2C2, R3C2, R3C1, by 1 cell per transition.

Image 1 positions
Arrow: R3C1

Image 2 positions
Arrow: R3C2

Solution - Exercise 14

Image 1
Matrix 3

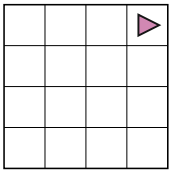
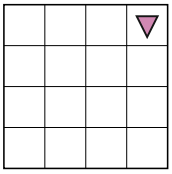


Image 2
Matrix 3



The triangle (initially pink) stays at R1C4; turns 90 degrees clockwise each time.

Image 1 positions
Triangle: R1C4

Image 2 positions
Triangle: R1C4

Solution - Exercise 15

Image 1
Matrix 1

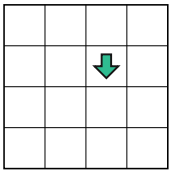


Image 2
Matrix 1

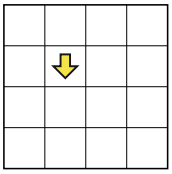


Image 1 positions
Arrow: R2C3

Image 2 positions
Arrow: R2C2

The arrow (initially green) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle green, yellow.

Solution - Exercise 16

Image 1
Matrix 2

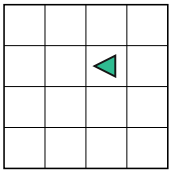


Image 2
Matrix 2

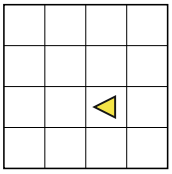


Image 1 positions
Triangle: R2C3

Image 2 positions
Triangle: R3C3

The triangle (initially green) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle green, yellow.

Solution - Exercise 17

Image 1
Matrix 1

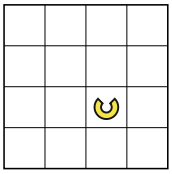
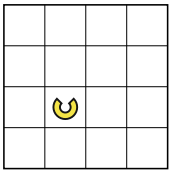


Image 2
Matrix 3



The crescent (initially yellow) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Crescent: R3C3

Image 2 positions
Crescent: R3C2

Solution - Exercise 18

Image 1
Matrix 2

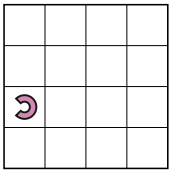
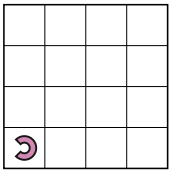


Image 2
Matrix 2



The crescent (initially pink) moves along column 1 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Crescent: R3C1

Image 2 positions
Crescent: R4C1

Solution - Exercise 19

Image 1
Matrix 3

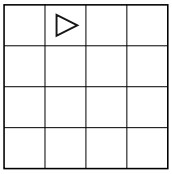
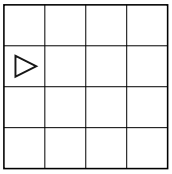


Image 2
Matrix 2



The triangle (initially white) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Triangle: R1C2

Image 2 positions
Triangle: R2C1

Solution - Exercise 20

Image 1
Matrix 3

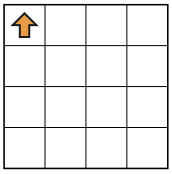
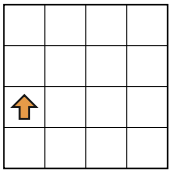


Image 2
Matrix 3



The arrow (initially orange) moves anticlockwise around the outer edge, by 2 cells per transition.

Image 1 positions
Arrow: R1C1

Image 2 positions
Arrow: R3C1

Solution - Exercise 21

Image 1
Matrix 3

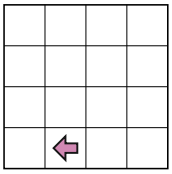
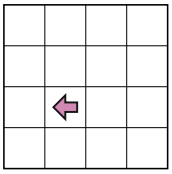


Image 2
Matrix 3



The arrow (initially pink) moves clockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition.

Image 1 positions
Arrow: R4C2

Image 2 positions
Arrow: R3C2

Solution - Exercise 22

Image 1
Matrix 2

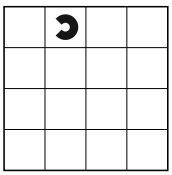
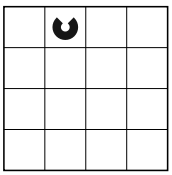


Image 2
Matrix 1



The crescent (initially black) stays at R1C2; turns 90 degrees clockwise each time.

Image 1 positions
Crescent: R1C2

Image 2 positions
Crescent: R1C2

Solution - Exercise 23

Image 1
Matrix 3

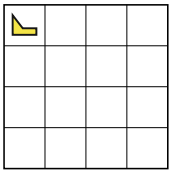
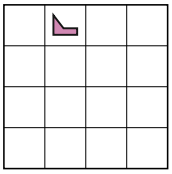


Image 2
Matrix 1



The flag (initially yellow) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle yellow, pink.

Image 1 positions
Flag: R1C1

Image 2 positions
Flag: R1C2

Solution - Exercise 24

Image 1
Matrix 1

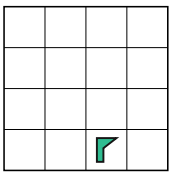
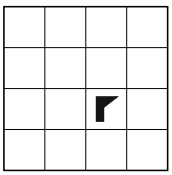


Image 2
Matrix 3



The flag (initially green) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle green, black.

Image 1 positions
Flag: R4C3

Image 2 positions
Flag: R3C3

Solution - Exercise 25

Image 1
Matrix 1

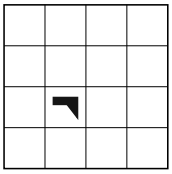
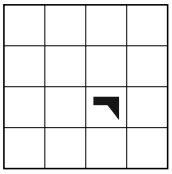


Image 2
Matrix 1



The flag (initially black) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Flag: R3C2

Image 2 positions
Flag: R3C3

Solution - Exercise 26

Image 1
Matrix 1

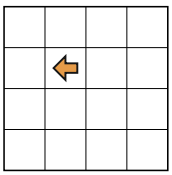
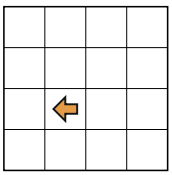


Image 2
Matrix 1



The arrow (initially orange) moves along column 2 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Arrow: R2C2

Image 2 positions
Arrow: R3C2

Solution - Exercise 27

Image 1
Matrix 2

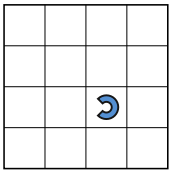
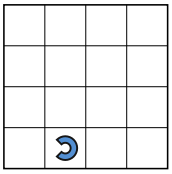


Image 2
Matrix 2



The crescent (initially blue) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Crescent: R3C3

Image 2 positions
Crescent: R4C2

Solution - Exercise 28

Image 1
Matrix 2

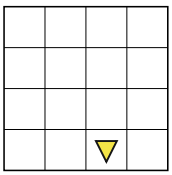
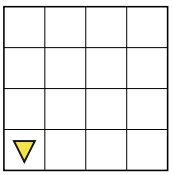


Image 2
Matrix 1



The triangle (initially yellow) moves clockwise around the outer edge, by 2 cells per transition.

Image 1 positions
Triangle: R4C3

Image 2 positions
Triangle: R4C1

Solution - Exercise 29

Image 1
Matrix 3

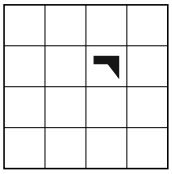
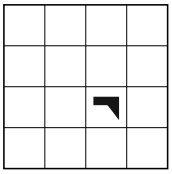


Image 2
Matrix 3



The flag (initially black) moves anticlockwise around the four cells R2C3, R2C4, R3C4, R3C3, by 1 cell per transition.

Image 1 positions
Flag: R2C3

Image 2 positions
Flag: R3C3

Solution - Exercise 30

Image 1
Matrix 3

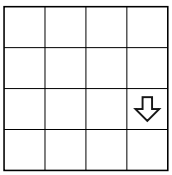
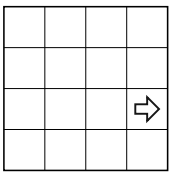


Image 2
Matrix 3



The arrow (initially white) stays at R3C4; turns 90 degrees anticlockwise each time.

Image 1 positions
Arrow: R3C4

Image 2 positions
Arrow: R3C4

Solution - Exercise 31

Image 1
Matrix 1

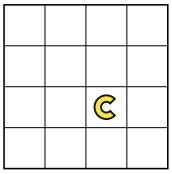
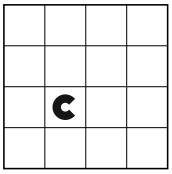


Image 2
Matrix 2



The crescent (initially yellow) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle yellow, black.

Image 1 positions
Crescent: R3C3

Image 2 positions
Crescent: R3C2

Solution - Exercise 32

Image 1
Matrix 3

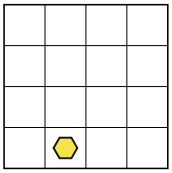
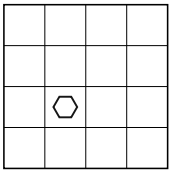


Image 2
Matrix 1



The hexagon (initially yellow) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle yellow, white.

Image 1 positions
Hexagon: R4C2

Image 2 positions
Hexagon: R3C2

Solution - Exercise 33

Image 1
Matrix 2

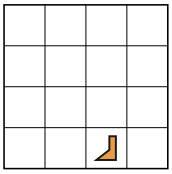
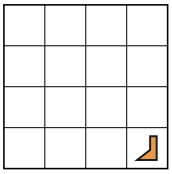


Image 2
Matrix 3



The flag (initially orange) moves along row 4 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Flag: R4C3

Image 2 positions
Flag: R4C4

Solution - Exercise 34

Image 1
Matrix 1

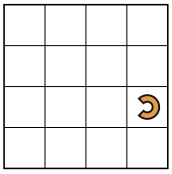
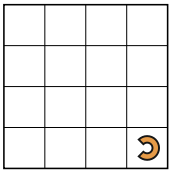


Image 2
Matrix 2



The crescent (initially orange) moves along column 4 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Crescent: R3C4

Image 2 positions
Crescent: R4C4

Solution - Exercise 35

Image 1
Matrix 3

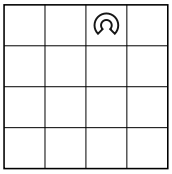
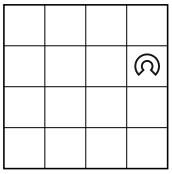


Image 2
Matrix 1



The crescent (initially white) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Crescent: R1C3

Image 2 positions
Crescent: R2C4

Solution - Exercise 36

Image 1
Matrix 2

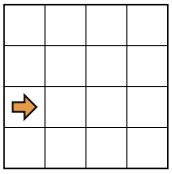
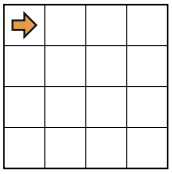


Image 2
Matrix 1



The arrow (initially orange) moves clockwise around the outer edge, by 2 cells per transition.

Image 1 positions
Arrow: R3C1

Image 2 positions
Arrow: R1C1

Solution - Exercise 37

Image 1
Matrix 2

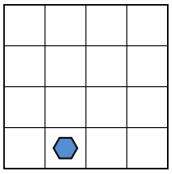
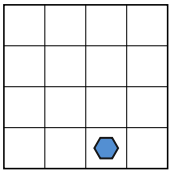


Image 2
Matrix 2



The hexagon (initially blue) moves anticlockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition.

Image 1 positions
Hexagon: R4C2

Image 2 positions
Hexagon: R4C3

Solution - Exercise 38

Image 1
Matrix 1

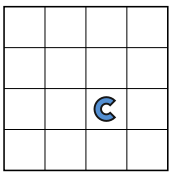
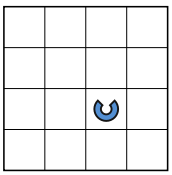


Image 2
Matrix 2



The crescent (initially blue) stays at R3C3; turns 90 degrees anticlockwise each time.

Image 1 positions
Crescent: R3C3

Image 2 positions
Crescent: R3C3

Solution - Exercise 39

Image 1
Matrix 3

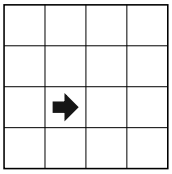


Image 2
Matrix 1

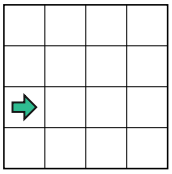


Image 1 positions
Arrow: R3C2

Image 2 positions
Arrow: R3C1

The arrow (initially black) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle black, green.

Solution - Exercise 40

Image 1
Matrix 2

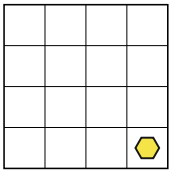


Image 2
Matrix 2

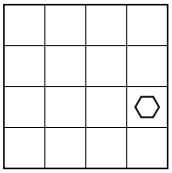


Image 1 positions
Hexagon: R4C4

Image 2 positions
Hexagon: R3C4

The hexagon (initially yellow) moves along column 4 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle yellow, white.

Solution - Exercise 41

Image 1
Matrix 2

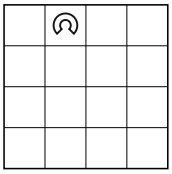
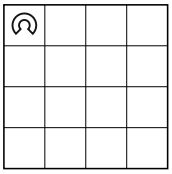


Image 2
Matrix 3



The crescent (initially white) moves along row 1 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Crescent: R1C2

Image 2 positions
Crescent: R1C1

Solution - Exercise 42

Image 1
Matrix 3

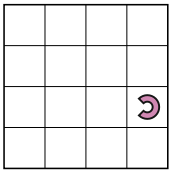
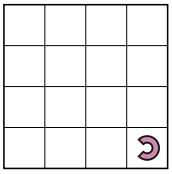


Image 2
Matrix 2



The crescent (initially pink) moves along column 4 by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Crescent: R3C4

Image 2 positions
Crescent: R4C4

Solution - Exercise 43

Image 1
Matrix 1

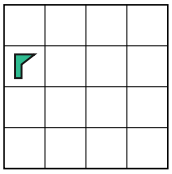
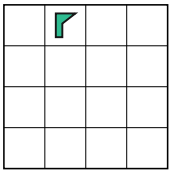


Image 2
Matrix 3



The flag (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

Image 1 positions
Flag: R2C1

Image 2 positions
Flag: R1C2

Solution - Exercise 44

Image 1
Matrix 2

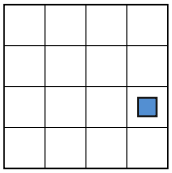
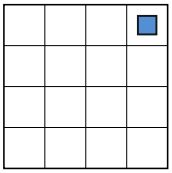


Image 2
Matrix 1



The square (initially blue) moves anticlockwise around the outer edge, by 2 cells per transition.

Image 1 positions
Square: R3C4

Image 2 positions
Square: R1C4

Solution - Exercise 45

Image 1
Matrix 1

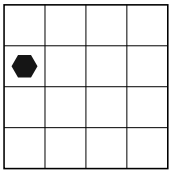
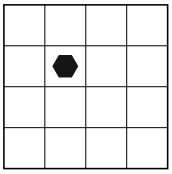


Image 2
Matrix 1



The hexagon (initially black) moves clockwise around the four cells R2C1, R2C2, R3C2, R3C1, by 1 cell per transition.

Image 1 positions
Hexagon: R2C1

Image 2 positions
Hexagon: R2C2

Solution - Exercise 46

Image 1
Matrix 1

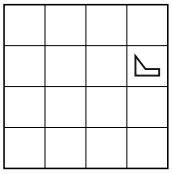
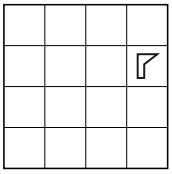


Image 2
Matrix 3



The flag (initially white) stays at R2C4; turns 90 degrees clockwise each time.

Image 1 positions
Flag: R2C4

Image 2 positions
Flag: R2C4

Solution - Exercise 47

Image 1
Matrix 1

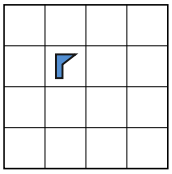


Image 2
Matrix 2

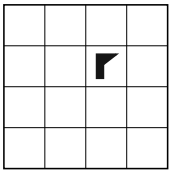


Image 1 positions
Flag: R2C2

Image 2 positions
Flag: R2C3

The flag (initially blue) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle blue, black.

Solution - Exercise 48

Image 1
Matrix 1

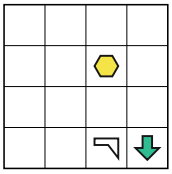


Image 2
Matrix 3

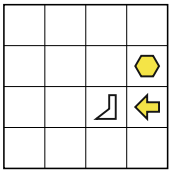


Image 1 positions
Arrow: R4C4
Hexagon: R2C3
Flag: R4C3

Image 2 positions
Arrow: R3C4
Hexagon: R2C4
Flag: R3C3

The arrow (initially green) moves along column 4 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle green, yellow.

The hexagon (initially yellow) moves along row 2 by 1 cell per transition, retracing the same line after each boundary.

The flag (initially white) moves clockwise around the four cells R3C3, R3C4, R4C4, R4C3, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 49

Image 1
Matrix 3

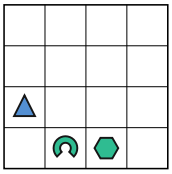


Image 2
Matrix 2

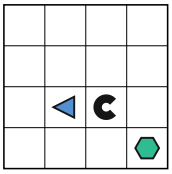


Image 1 positions
Crescent: R4C2
Hexagon: R4C3
Triangle: R3C1

Image 2 positions
Crescent: R3C3
Hexagon: R4C4
Triangle: R3C2

The crescent (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle green, black.

The hexagon (initially green) moves anticlockwise around the outer edge, by 1 cell per transition.

The triangle (initially blue) moves anticlockwise around the four cells R2C1, R2C2, R3C2, R3C1, by 1 cell per transition; turns 90 degrees anticlockwise each time.

Solution - Exercise 50

Image 1
Matrix 3

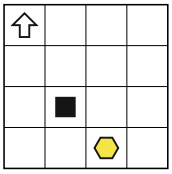


Image 2
Matrix 3

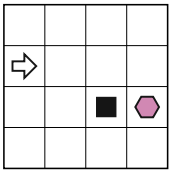


Image 1 positions
Hexagon: R4C3
Square: R3C2
Arrow: R1C1

Image 2 positions
Hexagon: R3C4
Square: R3C3
Arrow: R2C1

The hexagon (initially yellow) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle yellow, pink.

The square (initially black) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially white) moves anticlockwise around the outer edge, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 51

Image 1
Matrix 1

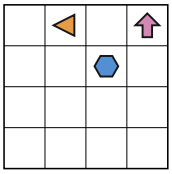


Image 2
Matrix 2

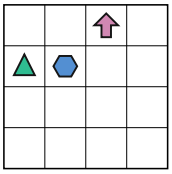


Image 1 positions
Triangle: R1C2
Arrow: R1C4
Hexagon: R2C3

Image 2 positions
Triangle: R2C1
Arrow: R1C3
Hexagon: R2C2

The triangle (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle orange, green.

The arrow (initially pink) moves anticlockwise around the outer edge, by 1 cell per transition.

The hexagon (initially blue) moves anticlockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition.

Solution - Exercise 52

Image 1
Matrix 2

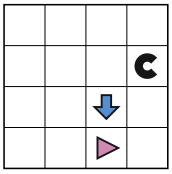


Image 2
Matrix 1

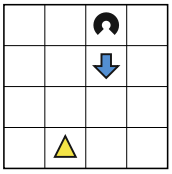


Image 1 positions
Triangle: R4C3
Arrow: R3C3
Crescent: R2C4

Image 2 positions
Triangle: R4C2
Arrow: R2C3
Crescent: R1C3

The triangle (initially pink) moves clockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle pink, yellow.

The arrow (initially blue) moves along column 3 by 1 cell per transition, retracing the same line after each boundary.

The crescent (initially black) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time.

Solution - Exercise 53

Image 1
Matrix 3

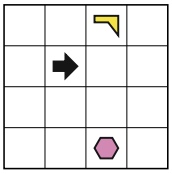


Image 2
Matrix 1

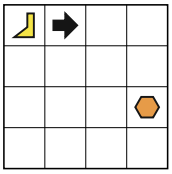


Image 1 positions
Hexagon: R4C3
Arrow: R2C2
Flag: R1C3

Image 2 positions
Hexagon: R3C4
Arrow: R1C2
Flag: R1C1

The hexagon (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle pink, orange.

The arrow (initially black) moves along column 2 by 1 cell per transition, retracing the same line after each boundary.

The flag (initially yellow) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time.

Solution - Exercise 54

Image 1
Matrix 2

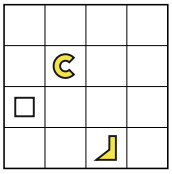


Image 2
Matrix 2

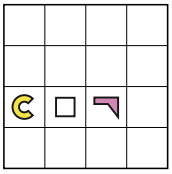


Image 1 positions
Flag: R4C3
Crescent: R2C2
Square: R3C1

Image 2 positions
Flag: R3C3
Crescent: R3C1
Square: R3C2

The flag (initially yellow) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle yellow, pink.

The crescent (initially yellow) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

The square (initially white) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 55

Image 1
Matrix 3

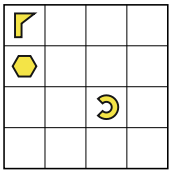


Image 2
Matrix 2

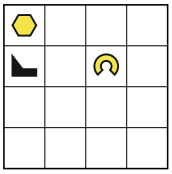


Image 1 positions
Flag: R1C1
Hexagon: R2C1
Crescent: R3C3

Image 2 positions
Flag: R2C1
Hexagon: R1C1
Crescent: R2C3

The flag (initially yellow) moves anticlockwise around the four cells R1C1, R1C2, R2C2, R2C1, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle yellow, black.

The hexagon (initially yellow) moves clockwise around the outer edge, by 1 cell per transition.

The crescent (initially yellow) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 56

Image 1
Matrix 2

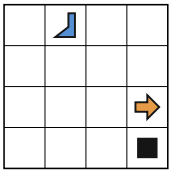


Image 2
Matrix 2

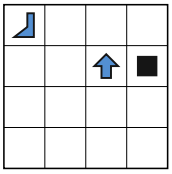


Image 1 positions
Arrow: R3C4
Flag: R1C2
Square: R4C4

Image 2 positions
Arrow: R2C3
Flag: R1C1
Square: R2C4

The arrow (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle orange, blue.

The flag (initially blue) moves along row 1 by 1 cell per transition, retracing the same line after each boundary.

The square (initially black) moves anticlockwise around the outer edge, by 2 cells per transition.

Solution - Exercise 57

Image 1
Matrix 2

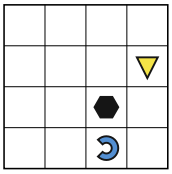


Image 2
Matrix 1

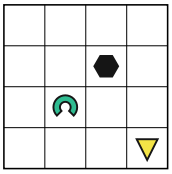


Image 1 positions
Crescent: R4C3
Triangle: R2C4
Hexagon: R3C3

Image 2 positions
Crescent: R3C2
Triangle: R4C4
Hexagon: R2C3

The crescent (initially blue) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle blue, green.

The triangle (initially yellow) moves clockwise around the outer edge, by 2 cells per transition.

The hexagon (initially black) moves along column 3 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 58

Image 1
Matrix 2

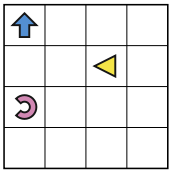


Image 2
Matrix 1

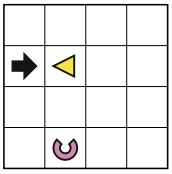


Image 1 positions
Arrow: R1C1
Triangle: R2C3
Crescent: R3C1

Image 2 positions
Arrow: R2C1
Triangle: R2C2
Crescent: R4C2

The arrow (initially blue) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle blue, black.

The triangle (initially yellow) moves along row 2 by 1 cell per transition, retracing the same line after each boundary.

The crescent (initially pink) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time.

Solution - Exercise 59

Image 1
Matrix 3

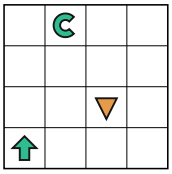


Image 2
Matrix 2

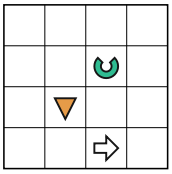


Image 1 positions
Arrow: R4C1
Triangle: R3C3
Crescent: R1C2

Image 2 positions
Arrow: R4C3
Triangle: R3C2
Crescent: R2C3

The arrow (initially green) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time; repeats the colour cycle green, white.

The triangle (initially orange) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

The crescent (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 60

Image 1
Matrix 1

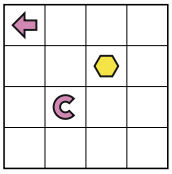


Image 2
Matrix 2

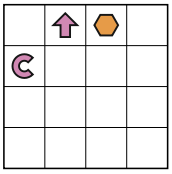


Image 1 positions
Hexagon: R2C3
Crescent: R3C2
Arrow: R1C1

Image 2 positions
Hexagon: R1C3
Crescent: R2C1
Arrow: R1C2

The hexagon (initially yellow) moves clockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition; repeats the colour cycle yellow, orange.

The crescent (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially pink) moves clockwise around the outer edge, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 61

Image 1
Matrix 3

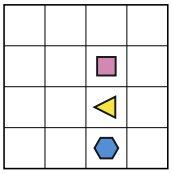


Image 2
Matrix 3

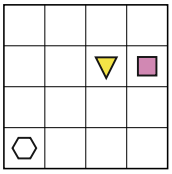


Image 1 positions
Hexagon: R4C3
Square: R2C3
Triangle: R3C3

Image 2 positions
Hexagon: R4C1
Square: R2C4
Triangle: R2C3

The hexagon (initially blue) moves clockwise around the outer edge, by 2 cells per transition; repeats the colour cycle blue, white.

The square (initially pink) moves along row 2 by 1 cell per transition, retracing the same line after each boundary.

The triangle (initially yellow) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 62

Image 1
Matrix 2

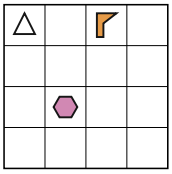


Image 2
Matrix 3

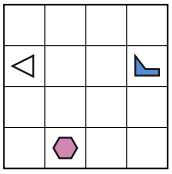


Image 1 positions
Flag: R1C3
Hexagon: R3C2
Triangle: R1C1

Image 2 positions
Flag: R2C4
Hexagon: R4C2
Triangle: R2C1

The flag (initially orange) moves clockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle orange, blue.

The hexagon (initially pink) moves along column 2 by 1 cell per transition, retracing the same line after each boundary.

The triangle (initially white) moves anticlockwise around the four cells R1C1, R1C2, R2C2, R2C1, by 1 cell per transition; turns 90 degrees anticlockwise each time.

Solution - Exercise 63

Image 1
Matrix 3

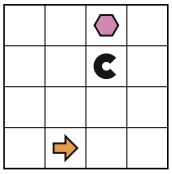


Image 2
Matrix 3

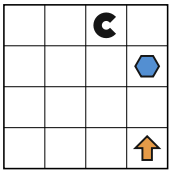


Image 1 positions
Hexagon: R1C3
Crescent: R2C3
Arrow: R4C2

Image 2 positions
Hexagon: R2C4
Crescent: R1C3
Arrow: R4C4

The hexagon (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle pink, blue.

The crescent (initially black) moves along column 3 by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially orange) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time.

Solution - Exercise 64

Image 1
Matrix 3

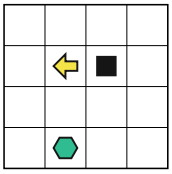


Image 2
Matrix 3

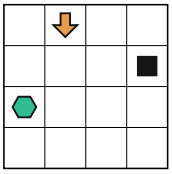


Image 1 positions
Arrow: R2C2
Hexagon: R4C2
Square: R2C3

Image 2 positions
Arrow: R1C2
Hexagon: R3C1
Square: R2C4

The arrow (initially yellow) moves clockwise around the four cells R1C2, R1C3, R2C3, R2C2, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle yellow, orange.

The hexagon (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

The square (initially black) moves along row 2 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 65

Image 1
Matrix 3

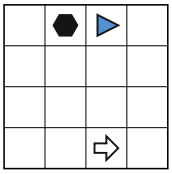


Image 2
Matrix 2

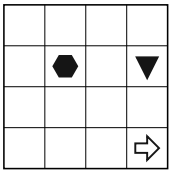


Image 1 positions
Triangle: R1C3
Arrow: R4C3
Hexagon: R1C2

Image 2 positions
Triangle: R2C4
Arrow: R4C4
Hexagon: R2C2

The triangle (initially blue) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle blue, black.

The arrow (initially white) moves anticlockwise around the outer edge, by 1 cell per transition.

The hexagon (initially black) moves clockwise around the four cells R1C1, R1C2, R2C2, R2C1, by 1 cell per transition.

Solution - Exercise 66

Image 1
Matrix 3

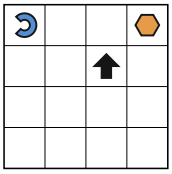


Image 2
Matrix 1

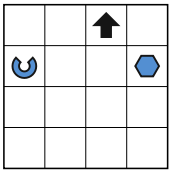


Image 1 positions
Hexagon: R1C4
Arrow: R2C3
Crescent: R1C1

Image 2 positions
Hexagon: R2C4
Arrow: R1C3
Crescent: R2C1

The hexagon (initially orange) moves clockwise around the outer edge, by 1 cell per transition; repeats the colour cycle orange, blue.

The arrow (initially black) moves along column 3 by 1 cell per transition, retracing the same line after each boundary.

The crescent (initially blue) moves anticlockwise around the four cells R1C1, R1C2, R2C2, R2C1, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 67

Image 1
Matrix 1

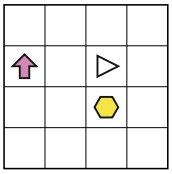


Image 2
Matrix 3

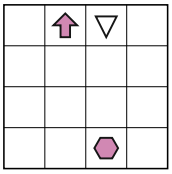


Image 1 positions
Hexagon: R3C3
Arrow: R2C1
Triangle: R2C3

Image 2 positions
Hexagon: R4C3
Arrow: R1C2
Triangle: R1C3

The hexagon (initially yellow) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle yellow, pink.

The arrow (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

The triangle (initially white) moves clockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 68

Image 1
Matrix 1

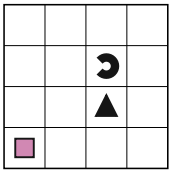


Image 2
Matrix 3

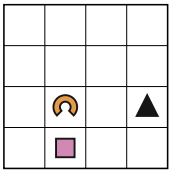


Image 1 positions
Crescent: R2C3
Triangle: R3C3
Square: R4C1

Image 2 positions
Crescent: R3C2
Triangle: R3C4
Square: R4C2

The crescent (initially black) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle black, orange.

The triangle (initially black) moves anticlockwise around the four cells R2C3, R2C4, R3C4, R3C3, by 1 cell per transition.

The square (initially pink) moves along row 4 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 69

Image 1
Matrix 2

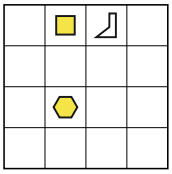


Image 2
Matrix 1

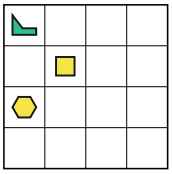


Image 1 positions
Flag: R1C3
Hexagon: R3C2
Square: R1C2

Image 2 positions
Flag: R1C1
Hexagon: R3C1
Square: R2C2

The flag (initially white) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time; repeats the colour cycle white, green.

The hexagon (initially yellow) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

The square (initially yellow) moves anticlockwise around the four cells R1C2, R1C3, R2C3, R2C2, by 1 cell per transition.

Solution - Exercise 70

Image 1
Matrix 2

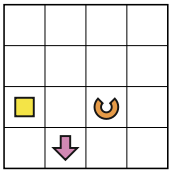


Image 2
Matrix 2

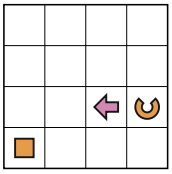


Image 1 positions
Square: R3C1
Crescent: R3C3
Arrow: R4C2

Image 2 positions
Square: R4C1
Crescent: R3C4
Arrow: R3C3

The square (initially yellow) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle yellow, orange.

The crescent (initially orange) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 71

Image 1
Matrix 1

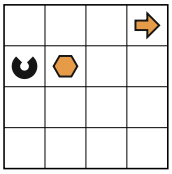


Image 2
Matrix 2

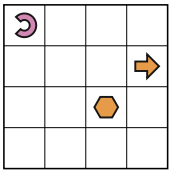


Image 1 positions
Crescent: R2C1
Arrow: R1C4
Hexagon: R2C2

Image 2 positions
Crescent: R1C1
Arrow: R2C4
Hexagon: R3C3

The crescent (initially black) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle black, pink.

The arrow (initially orange) moves clockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition.

The hexagon (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 72

Image 1
Matrix 3

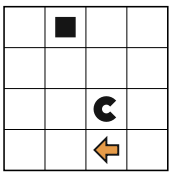


Image 2
Matrix 1

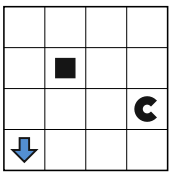


Image 1 positions
Arrow: R4C3
Crescent: R3C3
Square: R1C2

Image 2 positions
Arrow: R4C1
Crescent: R3C4
Square: R2C2

The arrow (initially orange) moves clockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle orange, blue.

The crescent (initially black) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

The square (initially black) moves along column 2 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 73

Image 1
Matrix 1

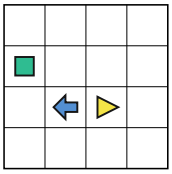


Image 2
Matrix 3

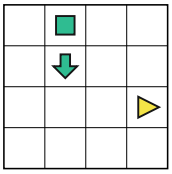


Image 1 positions
Arrow: R3C2
Triangle: R3C3
Square: R2C1

Image 2 positions
Arrow: R2C2
Triangle: R3C4
Square: R1C2

The arrow (initially blue) moves clockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle blue, green.

The triangle (initially yellow) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

The square (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 74

Image 1
Matrix 2

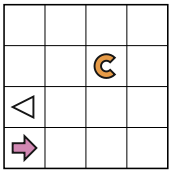


Image 2
Matrix 3

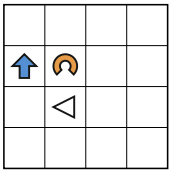


Image 1 positions
Arrow: R4C1
Triangle: R3C1
Crescent: R2C3

Image 2 positions
Arrow: R2C1
Triangle: R3C2
Crescent: R2C2

The arrow (initially pink) moves clockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle pink, blue.

The triangle (initially white) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

The crescent (initially orange) moves clockwise around the four cells R1C2, R1C3, R2C3, R2C2, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 75

Image 1
Matrix 3

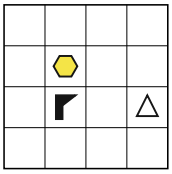


Image 2
Matrix 1

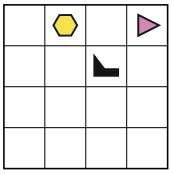


Image 1 positions
Triangle: R3C4
Hexagon: R2C2
Flag: R3C2

Image 2 positions
Triangle: R1C4
Hexagon: R1C2
Flag: R2C3

The triangle (initially white) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time; repeats the colour cycle white, pink.

The hexagon (initially yellow) moves along column 2 by 1 cell per transition, retracing the same line after each boundary.

The flag (initially black) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 76

Image 1
Matrix 1

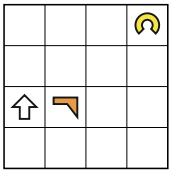


Image 2
Matrix 3

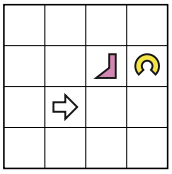


Image 1 positions
Flag: R3C2
Crescent: R1C4
Arrow: R3C1

Image 2 positions
Flag: R2C3
Crescent: R2C4
Arrow: R3C2

The flag (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle orange, pink.

The crescent (initially yellow) moves along column 4 by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially white) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 77

Image 1
Matrix 2

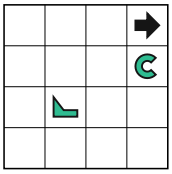


Image 2
Matrix 2

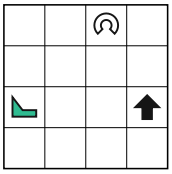


Image 1 positions
Crescent: R2C4
Flag: R3C2
Arrow: R1C4

Image 2 positions
Crescent: R1C3
Flag: R3C1
Arrow: R3C4

The crescent (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle green, white.

The flag (initially green) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially black) moves clockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time.

Solution - Exercise 78

Image 1
Matrix 3

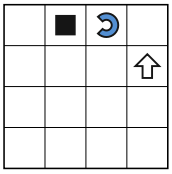


Image 2
Matrix 3

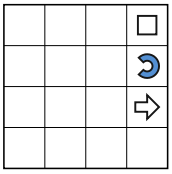


Image 1 positions
Square: R1C2
Crescent: R1C3
Arrow: R2C4

Image 2 positions
Square: R1C4
Crescent: R2C4
Arrow: R3C4

The square (initially black) moves clockwise around the outer edge, by 2 cells per transition; repeats the colour cycle black, white.

The crescent (initially blue) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially white) moves clockwise around the four cells R2C3, R2C4, R3C4, R3C3, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 79

Image 1
Matrix 1

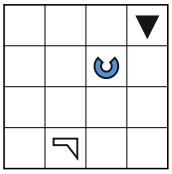


Image 2
Matrix 3

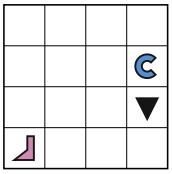


Image 1 positions
Flag: R4C2
Triangle: R1C4
Crescent: R2C3

Image 2 positions
Flag: R4C1
Triangle: R3C4
Crescent: R2C4

The flag (initially white) moves along row 4 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle white, pink.

The triangle (initially black) moves clockwise around the outer edge, by 2 cells per transition.

The crescent (initially blue) moves clockwise around the four cells R2C3, R2C4, R3C4, R3C3, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 80

Image 1
Matrix 3

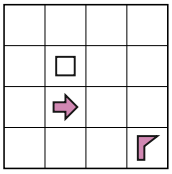


Image 2
Matrix 1

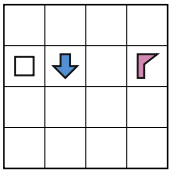


Image 1 positions
Arrow: R3C2
Flag: R4C4
Square: R2C2

Image 2 positions
Arrow: R2C2
Flag: R2C4
Square: R2C1

The arrow (initially pink) moves clockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition; turns 90 degrees clockwise each time; repeats the colour cycle pink, blue.

The flag (initially pink) moves anticlockwise around the outer edge, by 2 cells per transition.

The square (initially white) moves along row 2 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 81

Image 1
Matrix 2

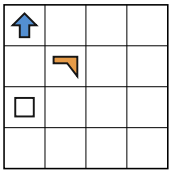


Image 2
Matrix 1

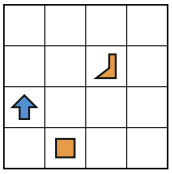


Image 1 positions
Square: R3C1
Arrow: R1C1
Flag: R2C2

Image 2 positions
Square: R4C2
Arrow: R3C1
Flag: R2C3

The square (initially white) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle white, orange.

The arrow (initially blue) moves anticlockwise around the outer edge, by 2 cells per transition.

The flag (initially orange) moves clockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 82

Image 1
Matrix 2

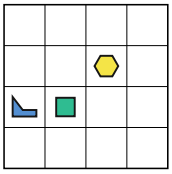


Image 2
Matrix 2

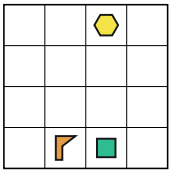


Image 1 positions
Flag: R3C1
Square: R3C2
Hexagon: R2C3

Image 2 positions
Flag: R4C2
Square: R4C3
Hexagon: R1C3

The flag (initially blue) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time; repeats the colour cycle blue, orange.

The square (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

The hexagon (initially yellow) moves along column 3 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 83

Image 1
Matrix 3

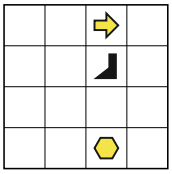


Image 2
Matrix 1

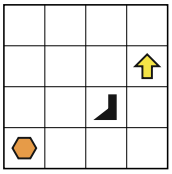


Image 1 positions
Hexagon: R4C3
Flag: R2C3
Arrow: R1C3

Image 2 positions
Hexagon: R4C1
Flag: R3C3
Arrow: R2C4

The hexagon (initially yellow) moves clockwise around the outer edge, by 2 cells per transition; repeats the colour cycle yellow, orange.

The flag (initially black) moves along column 3 by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially yellow) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 84

Image 1
Matrix 2

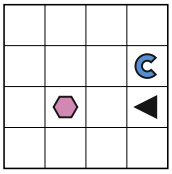


Image 2
Matrix 2

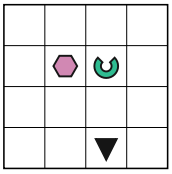


Image 1 positions
Crescent: R2C4
Hexagon: R3C2
Triangle: R3C4

Image 2 positions
Crescent: R2C3
Hexagon: R2C2
Triangle: R4C3

The crescent (initially blue) moves anticlockwise around the four cells R2C3, R2C4, R3C4, R3C3, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle blue, green.

The hexagon (initially pink) moves along column 2 by 1 cell per transition, retracing the same line after each boundary.

The triangle (initially black) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 85

Image 1
Matrix 1

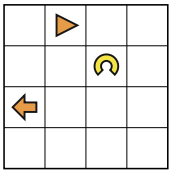


Image 2
Matrix 1

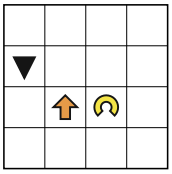


Image 1 positions
Triangle: R1C2
Crescent: R2C3
Arrow: R3C1

Image 2 positions
Triangle: R2C1
Crescent: R3C3
Arrow: R3C2

The triangle (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle orange, black.

The crescent (initially yellow) moves anticlockwise around the four cells R2C3, R2C4, R3C4, R3C3, by 1 cell per transition.

The arrow (initially orange) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 86

Image 1
Matrix 2

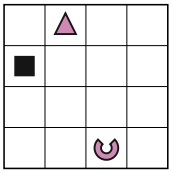


Image 2
Matrix 3

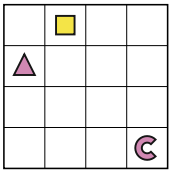


Image 1 positions
Square: R2C1
Triangle: R1C2
Crescent: R4C3

Image 2 positions
Square: R1C2
Triangle: R2C1
Crescent: R4C4

The square (initially black) moves clockwise around the outer edge, by 2 cells per transition; repeats the colour cycle black, yellow.

The triangle (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

The crescent (initially pink) moves along row 4 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 87

Image 1
Matrix 1

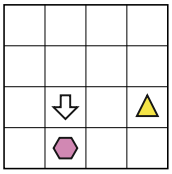


Image 2
Matrix 3

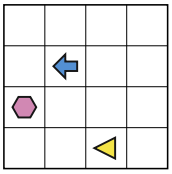


Image 1 positions
Arrow: R3C2
Hexagon: R4C2
Triangle: R3C4

Image 2 positions
Arrow: R2C2
Hexagon: R3C1
Triangle: R4C3

The arrow (initially white) moves anticlockwise around the four cells R2C1, R2C2, R3C2, R3C1, by 1 cell per transition; turns 90 degrees clockwise each time; repeats the colour cycle white, blue.

The hexagon (initially pink) moves clockwise around the outer edge, by 2 cells per transition.

The triangle (initially yellow) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 88

Image 1
Matrix 1

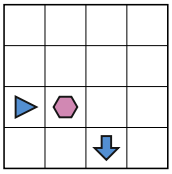


Image 2
Matrix 1

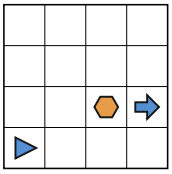


Image 1 positions
Hexagon: R3C2
Triangle: R3C1
Arrow: R4C3

Image 2 positions
Hexagon: R3C3
Triangle: R4C1
Arrow: R3C4

The hexagon (initially pink) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle pink, orange.

The triangle (initially blue) moves along column 1 by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially blue) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 89

Image 1
Matrix 1

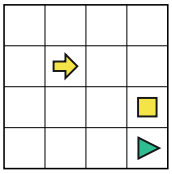


Image 2
Matrix 1

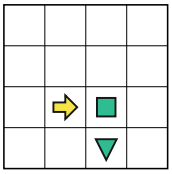


Image 1 positions
Square: R3C4
Arrow: R2C2
Triangle: R4C4

Image 2 positions
Square: R3C3
Arrow: R3C2
Triangle: R4C3

The square (initially yellow) moves anticlockwise around the four cells R3C3, R3C4, R4C4, R4C3, by 1 cell per transition; repeats the colour cycle yellow, green.

The arrow (initially yellow) moves along column 2 by 1 cell per transition, retracing the same line after each boundary.

The triangle (initially green) moves clockwise around the outer edge, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 90

Image 1
Matrix 2

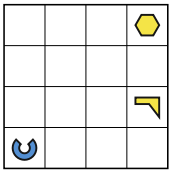


Image 2
Matrix 1

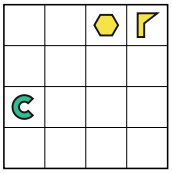


Image 1 positions
Crescent: R4C1
Hexagon: R1C4
Flag: R3C4

Image 2 positions
Crescent: R3C1
Hexagon: R1C3
Flag: R1C4

The crescent (initially blue) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle blue, green.

The hexagon (initially yellow) moves anticlockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition.

The flag (initially yellow) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time.

Solution - Exercise 91

Image 1
Matrix 1

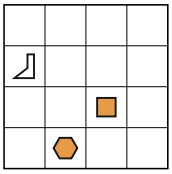


Image 2
Matrix 2

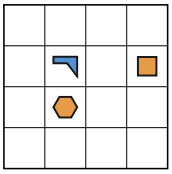


Image 1 positions
Flag: R2C1
Hexagon: R4C2
Square: R3C3

Image 2 positions
Flag: R2C2
Hexagon: R3C2
Square: R2C4

The flag (initially white) moves anticlockwise around the four cells R1C1, R1C2, R2C2, R2C1, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle white, blue.

The hexagon (initially orange) moves along column 2 by 1 cell per transition, retracing the same line after each boundary.

The square (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 92

Image 1
Matrix 1

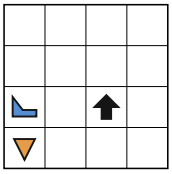


Image 2
Matrix 1

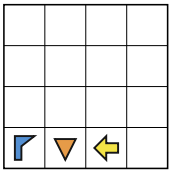


Image 1 positions
Arrow: R3C3
Triangle: R4C1
Flag: R3C1

Image 2 positions
Arrow: R4C3
Triangle: R4C2
Flag: R4C1

The arrow (initially black) moves clockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle black, yellow.

The triangle (initially orange) moves along row 4 by 1 cell per transition, retracing the same line after each boundary.

The flag (initially blue) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 93

Image 1
Matrix 1

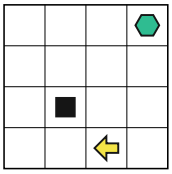


Image 2
Matrix 2

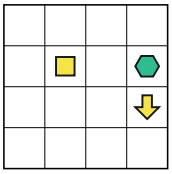


Image 1 positions
Square: R3C2
Hexagon: R1C4
Arrow: R4C3

Image 2 positions
Square: R2C2
Hexagon: R2C4
Arrow: R3C4

The square (initially black) moves anticlockwise around the four cells R2C1, R2C2, R3C2, R3C1, by 1 cell per transition; repeats the colour cycle black, yellow.

The hexagon (initially green) moves along column 4 by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially yellow) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 94

Image 1
Matrix 2

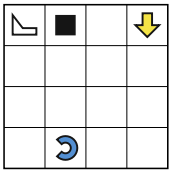


Image 2
Matrix 2

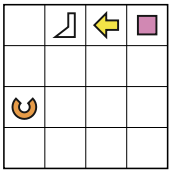


Image 1 positions
Square: R1C2
Flag: R1C1
Crescent: R4C2
Arrow: R1C4

Image 2 positions
Square: R1C4
Flag: R1C2
Crescent: R3C1
Arrow: R1C3

The square (initially white) moves clockwise around the outer edge, by 2 cells per transition; repeats the colour cycle white, black, pink.

The flag (initially white) moves along row 1 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

The crescent (initially blue) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle blue, orange.

The arrow (initially yellow) moves anticlockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 95

Image 1
Matrix 2

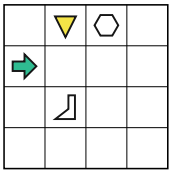


Image 2
Matrix 3

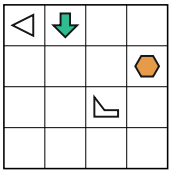


Image 1 positions
Hexagon: R1C3
Flag: R3C2
Triangle: R1C2
Arrow: R2C1

Image 2 positions
Hexagon: R2C4
Flag: R3C3
Triangle: R1C1
Arrow: R1C2

The hexagon (initially white) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle white, orange.

The flag (initially white) moves anticlockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions.

The triangle (initially yellow) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle yellow, white.

The arrow (initially green) moves clockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time.

Solution - Exercise 96

Image 1
Matrix 2

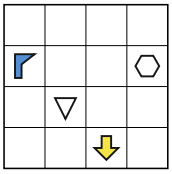


Image 2
Matrix 1

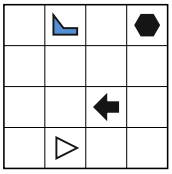


Image 1 positions
Arrow: R4C3
Triangle: R3C2
Hexagon: R2C4
Flag: R2C1

Image 2 positions
Arrow: R3C3
Triangle: R4C2
Hexagon: R1C4
Flag: R1C2

The arrow (initially yellow) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle yellow, black.

The triangle (initially white) moves anticlockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1, 2, 3, 4, then 5 cells on successive transitions; turns 90 degrees anticlockwise each time.

The hexagon (initially yellow) moves anticlockwise around the outer edge, by 1 cell per transition; repeats the colour cycle yellow, white, black.

The flag (initially blue) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 97

Image 1
Matrix 1

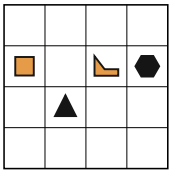


Image 2
Matrix 3

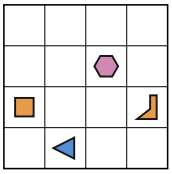


Image 1 positions
Triangle: R3C2
Square: R2C1
Hexagon: R2C4
Flag: R2C3

Image 2 positions
Triangle: R4C2
Square: R3C1
Hexagon: R2C3
Flag: R3C4

The triangle (initially pink) moves anticlockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle pink, black, blue.

The square (initially orange) moves along column 1 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary.

The hexagon (initially blue) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle blue, black, pink.

The flag (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 98

Image 1
Matrix 3

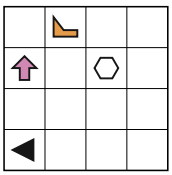


Image 2
Matrix 2

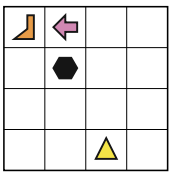


Image 1 positions
Hexagon: R2C3
Arrow: R2C1
Triangle: R4C1
Flag: R1C2

Image 2 positions
Hexagon: R2C2
Arrow: R1C2
Triangle: R4C3
Flag: R1C1

The hexagon (initially white) moves anticlockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition; repeats the colour cycle white, black.

The arrow (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions.

The triangle (initially black) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time; repeats the colour cycle black, yellow.

The flag (initially orange) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 99

Image 1
Matrix 1

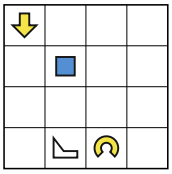


Image 2
Matrix 1

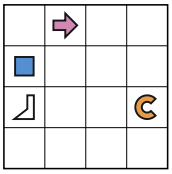


Image 1 positions
Arrow: R1C1
Square: R2C2
Crescent: R4C3
Flag: R4C2

Image 2 positions
Arrow: R1C2
Square: R2C1
Crescent: R3C4
Flag: R3C1

The arrow (initially black) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle black, yellow, pink.

The square (initially blue) moves anticlockwise around the four cells R2C1, R2C2, R3C2, R3C1, by 1, 2, 3, 4, then 5 cells on successive transitions.

The crescent (initially white) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle white, yellow, orange.

The flag (initially white) moves clockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time.

Solution - Exercise 100

Image 1
Matrix 3

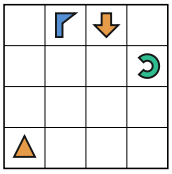


Image 2
Matrix 1

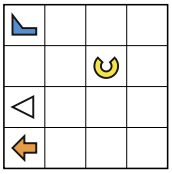


Image 1 positions
Triangle: R4C1
Arrow: R1C3
Crescent: R2C4
Flag: R1C2

Image 2 positions
Triangle: R3C1
Arrow: R4C1
Crescent: R2C3
Flag: R1C1

The triangle (initially orange) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle orange, white.

The arrow (initially orange) moves anticlockwise around the outer edge, by 1, 2, 3, 4, then 5 cells on successive transitions; turns 90 degrees clockwise each time.

The crescent (initially black) moves clockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition; turns 90 degrees clockwise each time; repeats the colour cycle black, green, yellow.

The flag (initially blue) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 101

Image 1
Matrix 3

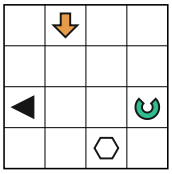


Image 2
Matrix 1

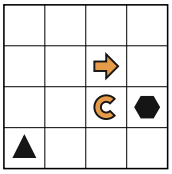


Image 1 positions
Crescent: R3C4
Triangle: R3C1
Hexagon: R4C3
Arrow: R1C2

Image 2 positions
Crescent: R3C3
Triangle: R4C1
Hexagon: R3C4
Arrow: R2C3

The crescent (initially blue) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle blue, green, orange.

The triangle (initially black) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

The hexagon (initially white) moves anticlockwise around the outer edge, by 2 cells per transition; repeats the colour cycle white, black.

The arrow (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 102

Image 1
Matrix 1

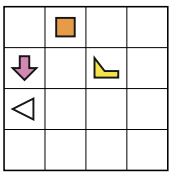


Image 2
Matrix 3

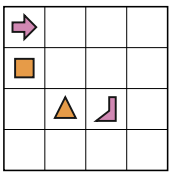


Image 1 positions
Flag: R2C3
Square: R1C2
Triangle: R3C1
Arrow: R2C1

Image 2 positions
Flag: R3C3
Square: R2C1
Triangle: R3C2
Arrow: R1C1

The flag (initially green) moves anticlockwise around the four cells R2C3, R2C4, R3C4, R3C3, by 1 cell per transition; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle green, yellow, pink.

The square (initially orange) moves along its diagonal by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary.

The triangle (initially green) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle green, white, orange.

The arrow (initially pink) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 103

Image 1
Matrix 2

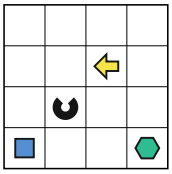


Image 2
Matrix 1

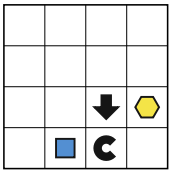


Image 1 positions
Hexagon: R4C4
Square: R4C1
Arrow: R2C3
Crescent: R3C2

Image 2 positions
Hexagon: R3C4
Square: R4C2
Arrow: R3C3
Crescent: R4C3

The hexagon (initially blue) moves along column 4 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle blue, green, yellow.

The square (initially blue) moves along row 4 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary.

The arrow (initially orange) moves anticlockwise around the four cells R2C3, R2C4, R3C4, R3C3, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle orange, yellow, black.

The crescent (initially black) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 104

Image 1
Matrix 1

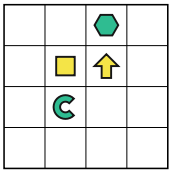


Image 2
Matrix 2

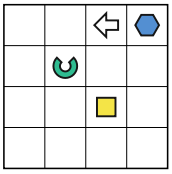


Image 1 positions
Hexagon: R1C3
Square: R2C2
Arrow: R2C3
Crescent: R3C2

Image 2 positions
Hexagon: R1C4
Square: R3C3
Arrow: R1C3
Crescent: R2C2

The hexagon (initially green) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle green, blue.

The square (initially yellow) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially yellow) moves anticlockwise around the four cells R1C2, R1C3, R2C3, R2C2, by 1 cell per transition; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle yellow, white.

The crescent (initially green) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 105

Image 1
Matrix 2

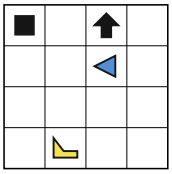


Image 2
Matrix 2

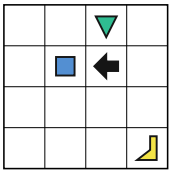


Image 1 positions
Triangle: R2C3
Arrow: R1C3
Square: R1C1
Flag: R4C2

Image 2 positions
Triangle: R1C3
Arrow: R2C3
Square: R2C2
Flag: R4C4

The triangle (initially blue) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle blue, green.

The arrow (initially black) moves clockwise around the four cells R1C2, R1C3, R2C3, R2C2, by 1, 2, 3, 4, then 5 cells on successive transitions; turns 90 degrees anticlockwise each time.

The square (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle green, black, blue.

The flag (initially yellow) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time.

Solution - Exercise 106

Image 1
Matrix 1

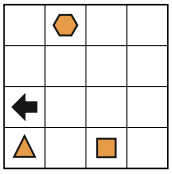


Image 2
Matrix 1

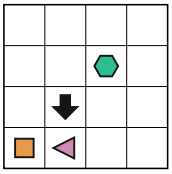


Image 1 positions
Hexagon: R1C2
Arrow: R3C1
Triangle: R4C1
Square: R4C3

Image 2 positions
Hexagon: R2C3
Arrow: R3C2
Triangle: R4C2
Square: R4C1

The hexagon (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle pink, orange, green.

The arrow (initially black) moves clockwise around the four cells R3C1, R3C2, R4C2, R4C1, by 1, 2, 3, 4, then 5 cells on successive transitions; turns 90 degrees anticlockwise each time.

The triangle (initially yellow) moves along row 4 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle yellow, orange, pink.

The square (initially orange) moves clockwise around the outer edge, by 2 cells per transition.

Solution - Exercise 107

Image 1
Matrix 3

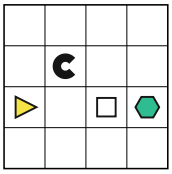


Image 2
Matrix 2

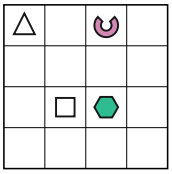


Image 1 positions
Triangle: R3C1
Hexagon: R3C4
Crescent: R2C2
Square: R3C3

Image 2 positions
Triangle: R1C1
Hexagon: R3C3
Crescent: R1C3
Square: R3C2

The triangle (initially yellow) moves clockwise around the outer edge, by 2 cells per transition; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle yellow, white.

The hexagon (initially green) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.

The crescent (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle green, black, pink.

The square (initially white) moves anticlockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition.

Solution - Exercise 108

Image 1
Matrix 3

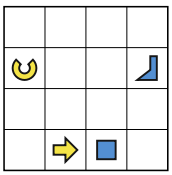


Image 2
Matrix 2

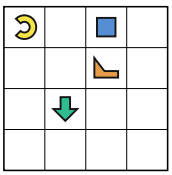


Image 1 positions
Flag: R2C4
Square: R4C3
Arrow: R4C2
Crescent: R2C1

Image 2 positions
Flag: R2C3
Square: R1C3
Arrow: R3C2
Crescent: R1C1

The flag (initially yellow) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle yellow, blue, orange.

The square (initially blue) moves anticlockwise around the outer edge, by 1, 2, 3, 4, then 5 cells on successive transitions.

The arrow (initially pink) moves clockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition; turns 90 degrees clockwise each time; repeats the colour cycle pink, yellow, green.

The crescent (initially yellow) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 109

Image 1
Matrix 2

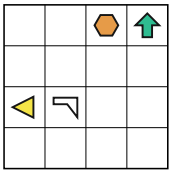


Image 2
Matrix 1

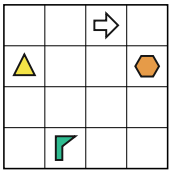


Image 1 positions
Arrow: R1C4
Hexagon: R1C3
Flag: R3C2
Triangle: R3C1

Image 2 positions
Arrow: R1C3
Hexagon: R2C4
Flag: R4C2
Triangle: R2C1

The arrow (initially green) moves anticlockwise around the outer edge, by 1 cell per transition; turns 90 degrees clockwise each time; repeats the colour cycle green, white.

The hexagon (initially orange) moves along its diagonal by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary.

The flag (initially white) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle white, green.

The triangle (initially yellow) moves clockwise around the four cells R2C1, R2C2, R3C2, R3C1, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 110

Image 1
Matrix 1

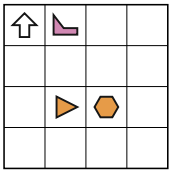


Image 2
Matrix 1

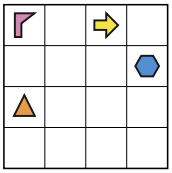


Image 1 positions
Hexagon: R3C3
Flag: R1C2
Arrow: R1C1
Triangle: R3C2

Image 2 positions
Hexagon: R2C4
Flag: R1C1
Arrow: R1C3
Triangle: R3C1

The hexagon (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle orange, blue.

The flag (initially pink) moves anticlockwise around the four cells R1C1, R1C2, R2C2, R2C1, by 1 cell per transition; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions.

The arrow (initially pink) moves clockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time; repeats the colour cycle pink, white, yellow.

The triangle (initially orange) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 111

Image 1
Matrix 1

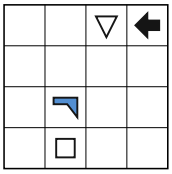


Image 2
Matrix 2

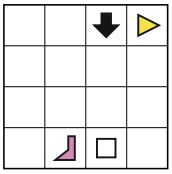


Image 1 positions
Flag: R3C2
Square: R4C2
Triangle: R1C3
Arrow: R1C4

Image 2 positions
Flag: R4C2
Square: R4C3
Triangle: R1C4
Arrow: R1C3

The flag (initially blue) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle blue, pink.

The square (initially white) moves along row 4 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary.

The triangle (initially white) moves clockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle white, yellow.

The arrow (initially black) moves anticlockwise around the outer edge, by 1 cell per transition; turns 90 degrees anticlockwise each time.

Solution - Exercise 112

Image 1
Matrix 2

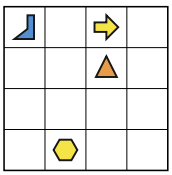


Image 2
Matrix 3

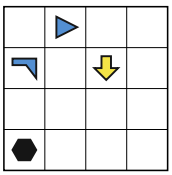


Image 1 positions
Hexagon: R4C2
Flag: R1C1
Triangle: R2C3
Arrow: R1C3

Image 2 positions
Hexagon: R4C1
Flag: R2C1
Triangle: R1C2
Arrow: R2C3

The hexagon (initially orange) moves along row 4 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle orange, yellow, black.

The flag (initially blue) moves along column 1 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

The triangle (initially white) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle white, orange, blue.

The arrow (initially yellow) moves anticlockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 113

Image 1
Matrix 3

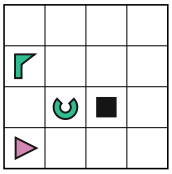


Image 2
Matrix 3

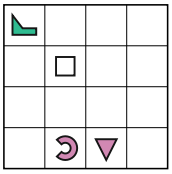


Image 1 positions
Crescent: R3C2
Flag: R2C1
Square: R3C3
Triangle: R4C1

Image 2 positions
Crescent: R4C2
Flag: R1C1
Square: R2C2
Triangle: R4C3

The crescent (initially black) moves anticlockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle black, green, pink.

The flag (initially green) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

The square (initially blue) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle blue, black, white.

The triangle (initially pink) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time.

Solution - Exercise 114

Image 1
Matrix 3

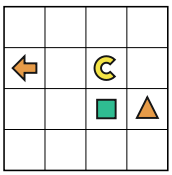


Image 2
Matrix 3

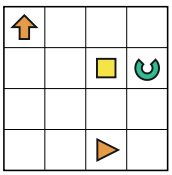


Image 1 positions
Crescent: R2C3
Triangle: R3C4
Square: R3C3
Arrow: R2C1

Image 2 positions
Crescent: R2C4
Triangle: R4C3
Square: R2C3
Arrow: R1C1

The crescent (initially yellow) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle yellow, green.

The triangle (initially orange) moves along its diagonal by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary; turns 90 degrees clockwise each time.

The square (initially white) moves anticlockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition; repeats the colour cycle white, green, yellow.

The arrow (initially orange) moves along column 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 115

Image 1
Matrix 2

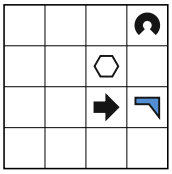


Image 2
Matrix 1

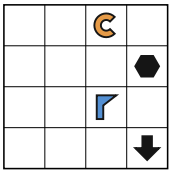


Image 1 positions
Hexagon: R2C3
Flag: R3C4
Crescent: R1C4
Arrow: R3C3

Image 2 positions
Hexagon: R2C4
Flag: R3C3
Crescent: R1C3
Arrow: R4C4

The hexagon (initially white) moves anticlockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition; repeats the colour cycle white, black.

The flag (initially blue) moves along row 3 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

The crescent (initially black) moves anticlockwise around the outer edge, by 1 cell per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle black, orange.

The arrow (initially black) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 116

Image 1
Matrix 2

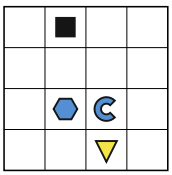


Image 2
Matrix 1

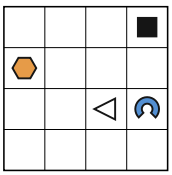


Image 1 positions
Triangle: R4C3
Square: R1C2
Hexagon: R3C2
Crescent: R3C3

Image 2 positions
Triangle: R3C3
Square: R1C4
Hexagon: R2C1
Crescent: R3C4

The triangle (initially blue) moves clockwise around the four cells R3C3, R3C4, R4C4, R4C3, by 1 cell per transition; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle blue, yellow, white.

The square (initially black) moves clockwise around the outer edge, by 2 cells per transition.

The hexagon (initially blue) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle blue, orange.

The crescent (initially blue) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 117

Image 1
Matrix 2

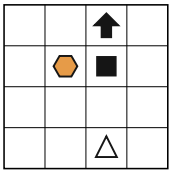


Image 2
Matrix 3

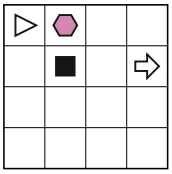


Image 1 positions
Arrow: R1C3
Triangle: R4C3
Hexagon: R2C2
Square: R2C3

Image 2 positions
Arrow: R2C4
Triangle: R1C1
Hexagon: R1C2
Square: R2C2

The arrow (initially green) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle green, black, white.

The triangle (initially white) moves clockwise around the outer edge, by 1, 2, 3, 4, then 5 cells on successive transitions; turns 90 degrees clockwise each time.

The hexagon (initially yellow) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle yellow, orange, pink.

The square (initially black) moves anticlockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition.

Solution - Exercise 118

Image 1
Matrix 3

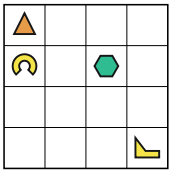


Image 2
Matrix 1

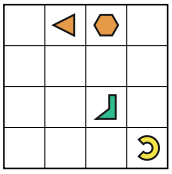


Image 1 positions
Flag: R4C4
Crescent: R2C1
Hexagon: R2C3
Triangle: R1C1

Image 2 positions
Flag: R3C3
Crescent: R4C4
Hexagon: R1C3
Triangle: R1C2

The flag (initially yellow) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle yellow, green.

The crescent (initially yellow) moves anticlockwise around the outer edge, by 1, 2, 3, 4, then 5 cells on successive transitions; turns 90 degrees clockwise each time.

The hexagon (initially black) moves anticlockwise around the four cells R1C2, R1C3, R2C3, R2C2, by 1 cell per transition; repeats the colour cycle black, green, orange.

The triangle (initially orange) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 119

Image 1
Matrix 1

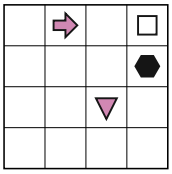


Image 2
Matrix 3

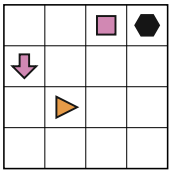


Image 1 positions
Triangle: R3C3
Hexagon: R2C4
Square: R1C4
Arrow: R1C2

Image 2 positions
Triangle: R3C2
Hexagon: R1C4
Square: R1C3
Arrow: R2C1

The triangle (initially green) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle green, pink, orange.

The hexagon (initially black) moves anticlockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition.

The square (initially black) moves anticlockwise around the outer edge, by 1 cell per transition; repeats the colour cycle black, white, pink.

The arrow (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 120

Image 1
Matrix 1

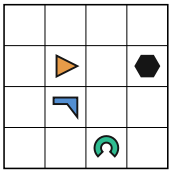


Image 2
Matrix 2

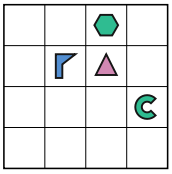


Image 1 positions
Hexagon: R2C4
Crescent: R4C3
Triangle: R2C2
Flag: R3C2

Image 2 positions
Hexagon: R1C3
Crescent: R3C4
Triangle: R2C3
Flag: R2C2

The hexagon (initially black) moves anticlockwise around the outer edge, by 2 cells per transition; repeats the colour cycle black, green.

The crescent (initially green) moves along its diagonal by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions.

The triangle (initially black) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle black, orange, pink.

The flag (initially blue) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 121

Image 1
Matrix 3

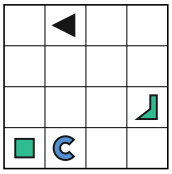


Image 2
Matrix 3

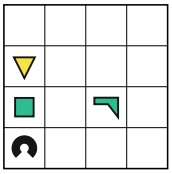


Image 1 positions
Crescent: R4C2
Flag: R3C4
Triangle: R1C2
Square: R4C1

Image 2 positions
Crescent: R4C1
Flag: R3C3
Triangle: R2C1
Square: R3C1

The crescent (initially blue) moves along row 4 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle blue, black.

The flag (initially green) moves anticlockwise around the four cells R3C3, R3C4, R4C4, R4C3, by 1, 2, 3, 4, then 5 cells on successive transitions; turns 90 degrees anticlockwise each time.

The triangle (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle pink, black, yellow.

The square (initially green) moves along column 1 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 122

Image 1
Matrix 2

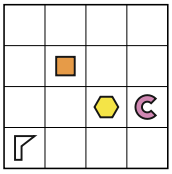


Image 2
Matrix 2

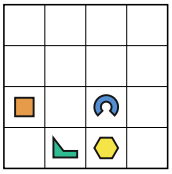


Image 1 positions
Crescent: R3C4
Square: R2C2
Flag: R4C1
Hexagon: R3C3

Image 2 positions
Crescent: R3C3
Square: R3C1
Flag: R4C2
Hexagon: R4C3

The crescent (initially pink) moves clockwise around the four cells R2C3, R2C4, R3C4, R3C3, by 1 cell per transition; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle pink, blue.

The square (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary.

The flag (initially pink) moves along row 4 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle pink, white, green.

The hexagon (initially yellow) moves along column 3 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 123

Image 1
Matrix 3

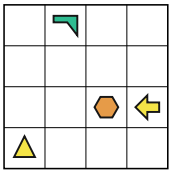


Image 2
Matrix 2

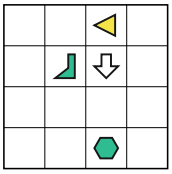


Image 1 positions
Arrow: R3C4
Triangle: R4C1
Hexagon: R3C3
Flag: R1C2

Image 2 positions
Arrow: R2C3
Triangle: R1C3
Hexagon: R4C3
Flag: R2C2

The arrow (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle pink, yellow, white.

The triangle (initially yellow) moves clockwise around the outer edge, by 1, 2, 3, 4, then 5 cells on successive transitions; turns 90 degrees anticlockwise each time.

The hexagon (initially white) moves clockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition; repeats the colour cycle white, orange, green.

The flag (initially green) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 124

Image 1
Matrix 1

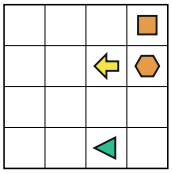


Image 2
Matrix 3

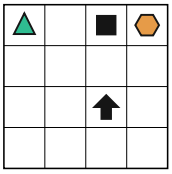


Image 1 positions
Arrow: R2C3
Triangle: R4C3
Square: R1C4
Hexagon: R2C4

Image 2 positions
Arrow: R3C3
Triangle: R1C1
Square: R1C3
Hexagon: R1C4

The arrow (initially yellow) moves clockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition; turns 90 degrees clockwise each time; repeats the colour cycle yellow, black.

The triangle (initially green) moves clockwise around the outer edge, by 1, 2, 3, 4, then 5 cells on successive transitions; turns 90 degrees clockwise each time.

The square (initially orange) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle orange, black.

The hexagon (initially orange) moves along column 4 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 125

Image 1
Matrix 2

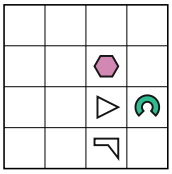


Image 2
Matrix 2

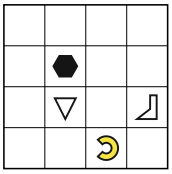


Image 1 positions
Crescent: R3C4
Triangle: R3C3
Hexagon: R2C3
Flag: R4C3

Image 2 positions
Crescent: R4C3
Triangle: R3C2
Hexagon: R2C2
Flag: R3C4

The crescent (initially white) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle white, green, yellow.

The triangle (initially white) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

The hexagon (initially pink) moves clockwise around the four cells R1C2, R1C3, R2C3, R2C2, by 1 cell per transition; repeats the colour cycle pink, black.

The flag (initially white) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees clockwise each time.

Solution - Exercise 126

Image 1
Matrix 2

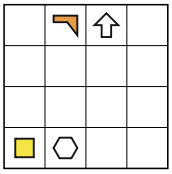


Image 2
Matrix 3

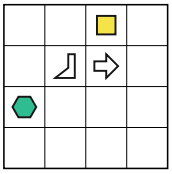


Image 1 positions
Hexagon: R4C2
Square: R4C1
Flag: R1C2
Arrow: R1C3

Image 2 positions
Hexagon: R3C1
Square: R1C3
Flag: R2C2
Arrow: R2C3

The hexagon (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle pink, white, green.

The square (initially yellow) moves clockwise around the outer edge, by 1, 2, 3, 4, then 5 cells on successive transitions.

The flag (initially green) moves clockwise around the four cells R1C1, R1C2, R2C2, R2C1, by 1 cell per transition; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle green, orange, white.

The arrow (initially white) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 127

Image 1
Matrix 3

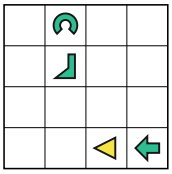


Image 2
Matrix 2

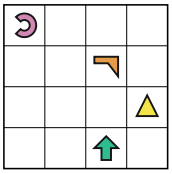


Image 1 positions
Crescent: R1C2
Triangle: R4C3
Flag: R2C2
Arrow: R4C4

Image 2 positions
Crescent: R1C1
Triangle: R3C4
Flag: R2C3
Arrow: R4C3

The crescent (initially yellow) moves anticlockwise around the outer edge, by 1 cell per transition; turns 90 degrees clockwise each time; repeats the colour cycle yellow, green, pink.

The triangle (initially yellow) moves along its diagonal by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary; turns 90 degrees clockwise each time.

The flag (initially green) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle green, orange.

The arrow (initially green) moves clockwise around the four cells R3C3, R3C4, R4C4, R4C3, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 128

Image 1
Matrix 3

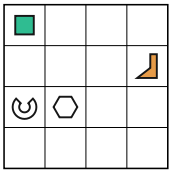


Image 2
Matrix 1

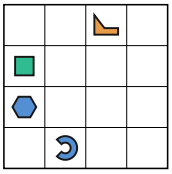


Image 1 positions
Hexagon: R3C2
Flag: R2C4
Crescent: R3C1
Square: R1C1

Image 2 positions
Hexagon: R3C1
Flag: R1C3
Crescent: R4C2
Square: R2C1

The hexagon (initially white) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle white, blue.

The flag (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions.

The crescent (initially white) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle white, blue.

The square (initially green) moves along column 1 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 129

Image 1
Matrix 1

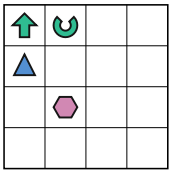


Image 2
Matrix 2

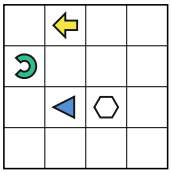


Image 1 positions
Hexagon: R3C2
Triangle: R2C1
Arrow: R1C1
Crescent: R1C2

Image 2 positions
Hexagon: R3C3
Triangle: R3C2
Arrow: R1C2
Crescent: R2C1

The hexagon (initially pink) moves anticlockwise around the four cells R2C2, R2C3, R3C3, R3C2, by 1 cell per transition; repeats the colour cycle pink, white.

The triangle (initially blue) moves along its diagonal by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions.

The arrow (initially black) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle black, green, yellow.

The crescent (initially green) moves anticlockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time.

Solution - Exercise 130

Image 1
Matrix 3

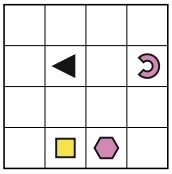


Image 2
Matrix 2

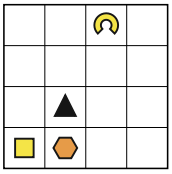


Image 1 positions
Crescent: R2C4
Square: R4C2
Hexagon: R4C3
Triangle: R2C2

Image 2 positions
Crescent: R1C3
Square: R4C1
Hexagon: R4C2
Triangle: R3C2

The crescent (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time; repeats the colour cycle pink, yellow.

The square (initially yellow) moves along row 4 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary.

The hexagon (initially pink) moves clockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition; repeats the colour cycle pink, orange.

The triangle (initially black) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 131

Image 1
Matrix 2

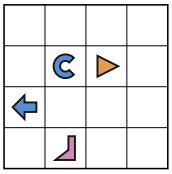


Image 2
Matrix 1

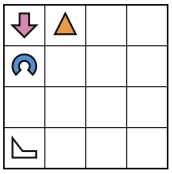


Image 1 positions
Flag: R4C2
Crescent: R2C2
Arrow: R3C1
Triangle: R2C3

Image 2 positions
Flag: R4C1
Crescent: R2C1
Arrow: R1C1
Triangle: R1C2

The flag (initially orange) moves clockwise around the four cells R3C1, R3C2, R4C2, R4C1, by 1 cell per transition; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle orange, pink, white.

The crescent (initially blue) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

The arrow (initially white) moves clockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle white, blue, pink.

The triangle (initially orange) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

Solution - Exercise 132

Image 1
Matrix 2

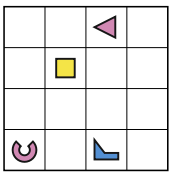


Image 2
Matrix 1

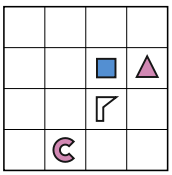


Image 1 positions
Flag: R4C3
Triangle: R1C3
Square: R2C2
Crescent: R4C1

Image 2 positions
Flag: R3C3
Triangle: R2C4
Square: R2C3
Crescent: R4C2

The flag (initially blue) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle blue, white.

The triangle (initially pink) moves along its diagonal by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary; turns 90 degrees clockwise each time.

The square (initially green) moves along row 2 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle green, yellow, blue.

The crescent (initially pink) moves anticlockwise around the outer edge, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 133

Image 1
Matrix 1

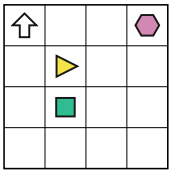


Image 2
Matrix 2

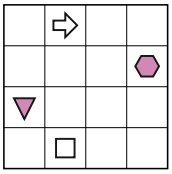


Image 1 positions
Square: R3C2
Hexagon: R1C4
Triangle: R2C2
Arrow: R1C1

Image 2 positions
Square: R4C2
Hexagon: R2C4
Triangle: R3C1
Arrow: R1C2

The square (initially green) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle green, white.

The hexagon (initially pink) moves clockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1, 2, 3, 4, then 5 cells on successive transitions.

The triangle (initially yellow) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle yellow, pink.

The arrow (initially white) moves along row 1 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 134

Image 1
Matrix 3

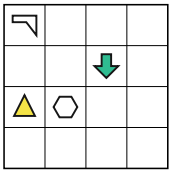


Image 2
Matrix 3

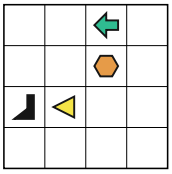


Image 1 positions
Flag: R1C1
Triangle: R3C1
Hexagon: R3C2
Arrow: R2C3

Image 2 positions
Flag: R3C1
Triangle: R3C2
Hexagon: R2C3
Arrow: R1C3

The flag (initially white) moves anticlockwise around the outer edge, by 2 cells per transition; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle white, black.

The triangle (initially yellow) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

The hexagon (initially white) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle white, orange.

The arrow (initially green) moves along column 3 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time.

Solution - Exercise 135

Image 1
Matrix 1

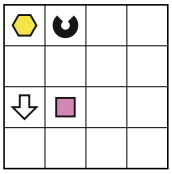


Image 2
Matrix 3

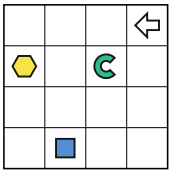


Image 1 positions
Square: R3C2
Arrow: R3C1
Crescent: R1C2
Hexagon: R1C1

Image 2 positions
Square: R4C2
Arrow: R1C4
Crescent: R2C3
Hexagon: R2C1

The square (initially orange) moves anticlockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition; repeats the colour cycle orange, pink, blue.

The arrow (initially white) moves clockwise around the outer edge, by 1, 2, 3, 4, then 5 cells on successive transitions; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions.

The crescent (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle pink, black, green.

The hexagon (initially yellow) moves along column 1 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 136

Image 1
Matrix 1

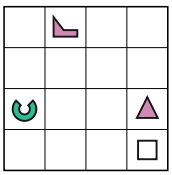


Image 2
Matrix 3

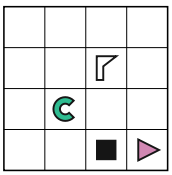


Image 1 positions
Square: R4C4
Triangle: R3C4
Flag: R1C2
Crescent: R3C1

Image 2 positions
Square: R4C3
Triangle: R4C4
Flag: R2C3
Crescent: R3C2

The square (initially blue) moves along row 4 by 1 cell per transition, retracing the same line after each boundary; repeats the colour cycle blue, white, black.

The triangle (initially pink) moves along column 4 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary; turns 90 degrees clockwise each time.

The flag (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle pink, white.

The crescent (initially green) moves anticlockwise around the four cells R2C1, R2C2, R3C2, R3C1, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 137

Image 1
Matrix 1

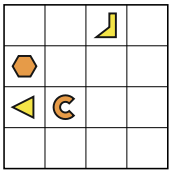


Image 2
Matrix 3

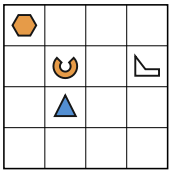


Image 1 positions
Triangle: R3C1
Crescent: R3C2
Flag: R1C3
Hexagon: R2C1

Image 2 positions
Triangle: R3C2
Crescent: R2C2
Flag: R2C4
Hexagon: R1C1

The triangle (initially white) moves along row 3 by 1 cell per transition, retracing the same line after each boundary; turns clockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle white, yellow, blue.

The crescent (initially orange) moves along column 2 by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees anticlockwise each time.

The flag (initially black) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle black, yellow, white.

The hexagon (initially orange) moves clockwise around the four cells R1C1, R1C2, R2C2, R2C1, by 1 cell per transition.

Solution - Exercise 138

Image 1
Matrix 3

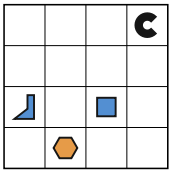


Image 2
Matrix 1

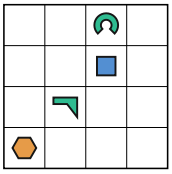


Image 1 positions
Flag: R3C1
Hexagon: R4C2
Crescent: R1C4
Square: R3C3

Image 2 positions
Flag: R3C2
Hexagon: R4C1
Crescent: R1C3
Square: R2C3

The flag (initially blue) moves anticlockwise around the four cells R2C1, R2C2, R3C2, R3C1, by 1 cell per transition; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle blue, green.

The hexagon (initially orange) moves along row 4 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary.

The crescent (initially pink) moves anticlockwise around the outer edge, by 1 cell per transition; turns 90 degrees clockwise each time; repeats the colour cycle pink, black, green.

The square (initially blue) moves along column 3 by 1 cell per transition, retracing the same line after each boundary.

Solution - Exercise 139

Image 1
Matrix 1

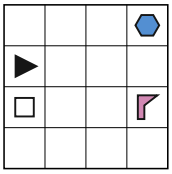


Image 2
Matrix 1

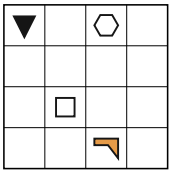


Image 1 positions
Flag: R3C4
Square: R3C1
Hexagon: R1C4
Triangle: R2C1

Image 2 positions
Flag: R4C3
Square: R3C2
Hexagon: R1C3
Triangle: R1C1

The flag (initially pink) moves along its diagonal by 1 cell per transition, retracing the same line after each boundary; turns 90 degrees clockwise each time; repeats the colour cycle pink, orange.

The square (initially white) moves along row 3 by 1, 2, 3, 4, then 5 cells on successive transitions, retracing the same line after each boundary.

The hexagon (initially blue) moves anticlockwise around the four cells R1C3, R1C4, R2C4, R2C3, by 1 cell per transition; repeats the colour cycle blue, white.

The triangle (initially black) moves clockwise around the outer edge, by 1 cell per transition; turns 90 degrees clockwise each time.

Solution - Exercise 140

Image 1
Matrix 2

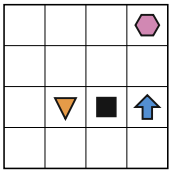


Image 2
Matrix 2

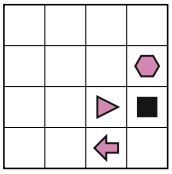


Image 1 positions
Triangle: R3C2
Hexagon: R1C4
Arrow: R3C4
Square: R3C3

Image 2 positions
Triangle: R3C3
Hexagon: R2C4
Arrow: R4C3
Square: R3C4

The triangle (initially white) moves clockwise around the four cells R3C2, R3C3, R4C3, R4C2, by 1 cell per transition; turns anticlockwise by 90, 180, 270, 360, then 450 degrees on successive transitions; repeats the colour cycle white, orange, pink.

The hexagon (initially pink) moves along column 4 by 1 cell per transition, retracing the same line after each boundary.

The arrow (initially blue) moves clockwise around the outer edge, by 2 cells per transition; turns 90 degrees anticlockwise each time; repeats the colour cycle blue, pink.

The square (initially black) moves along row 3 by 1 cell per transition, retracing the same line after each boundary.